

WEEK 2 – TESTING DOCUMENTATION

Educational Microcontroller Simulator – STC89C52

1. Testing Objective

The objective of testing is to verify that every selected instruction executes correctly, that the CPU state changes as expected, and that the complete demonstration program executes and terminates correctly.

2. Testing Method

Each selected instruction was tested individually using a small program in the simulator UI. The program was loaded, the simulator was reset, and execution was observed using Step. The complete demonstration program was then tested to verify that the supported instructions work together and that the program terminates correctly.

3. Individual Instruction Test Cases and Results

Test ID	Instruction	Test Program / Input	Expected Result	Actual Result	Status
TC01	MOV	MOV A,#10 HALT	A = 0AH	A = 0AH	PASS
TC02	MOVC	MOVC A,10 HALT	A = 03H	A = 03H	PASS
TC03	ADD	MOV A,#10 ADD A,#5 HALT	A = 0FH	A = 0FH	PASS
TC04	SUB	MOV A,#10 SUB A,#2 HALT	A = 08H	A = 08H	PASS
TC05	ANL	MOV A,#10 ANL A,#3 HALT	A = 02H	A = 02H	PASS
TC06	INC	MOV A,#10 INC A HALT	A = 0BH	A = 0BH	PASS
TC07	SJMP	MOV A,#10 SJMP 3 INC A HALT	PC = 03H; INC skipped	PC = 03H; INC skipped	PASS
TC08	HALT	MOV A,#10 HALT	Execution stops	Status = HALTED	PASS

4. Complete Demonstration Program Test

Demonstration Program:

```
MOV A, #10
MOVC A, 10
ADD A, #5
SUB A, #2
ANL A, #3
INC A
SJMP 7
HALT
```

Test ID	Expected Result	Actual Result	Status
TC09	A = 03H; PC = 07H; execution terminated	A = 03H; PC = 07H; status = HALTED	PASS

5. Overall Testing Result

All 8 selected instructions were individually tested and passed. The complete demonstration program was also tested successfully and terminated as expected. The Week 2 testing requirements were therefore satisfied for the implemented instruction set and demonstration program.

6. Testing Summary

Testing Item	Result
Individual instruction tests	8 / 8 PASS
Complete demonstration program	PASS
Overall testing result	PASS

This document follows the Week 2 requirement that every selected instruction has at least one test case and that both individual instructions and the complete demonstration program are tested.